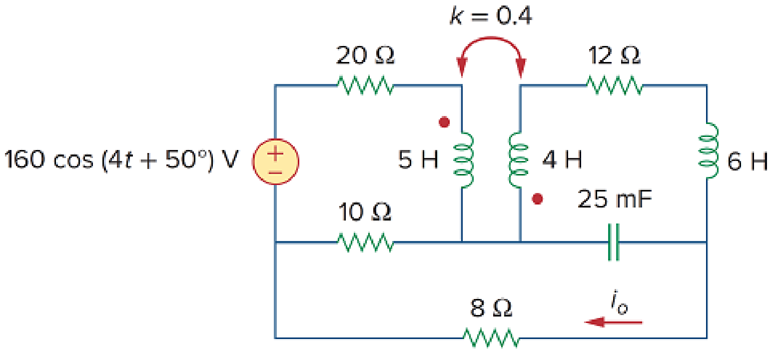
Chapter 13, Problem 13PP

Problem

Find *io* in the circuit of Fig. 13.55, using *PSpice*.

Figure 13.55



Step-by-step solution

Step 1 of 6

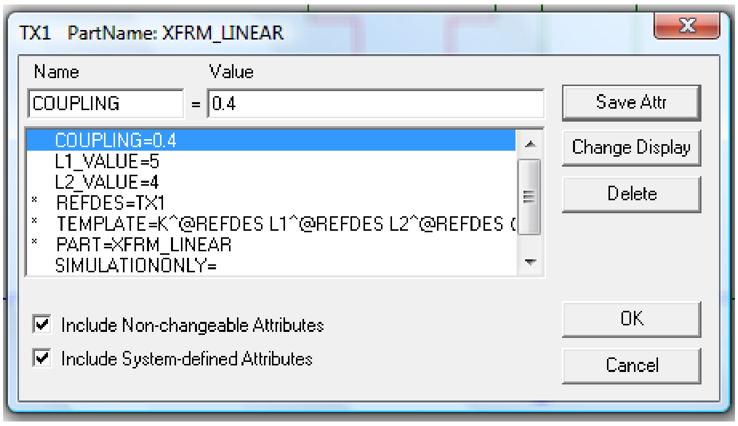
Step 1:

Select draw / get new part and type XFRM_LINEAR.

Click ok and place the XFRM_LINEAR symbol on the schematic, as shown in below figures (2) and double click on XFRM_LINERR device and set coupling = 0.4,

L_1_value $= 5$ and L_2 _ value $= 4$ as shown below.

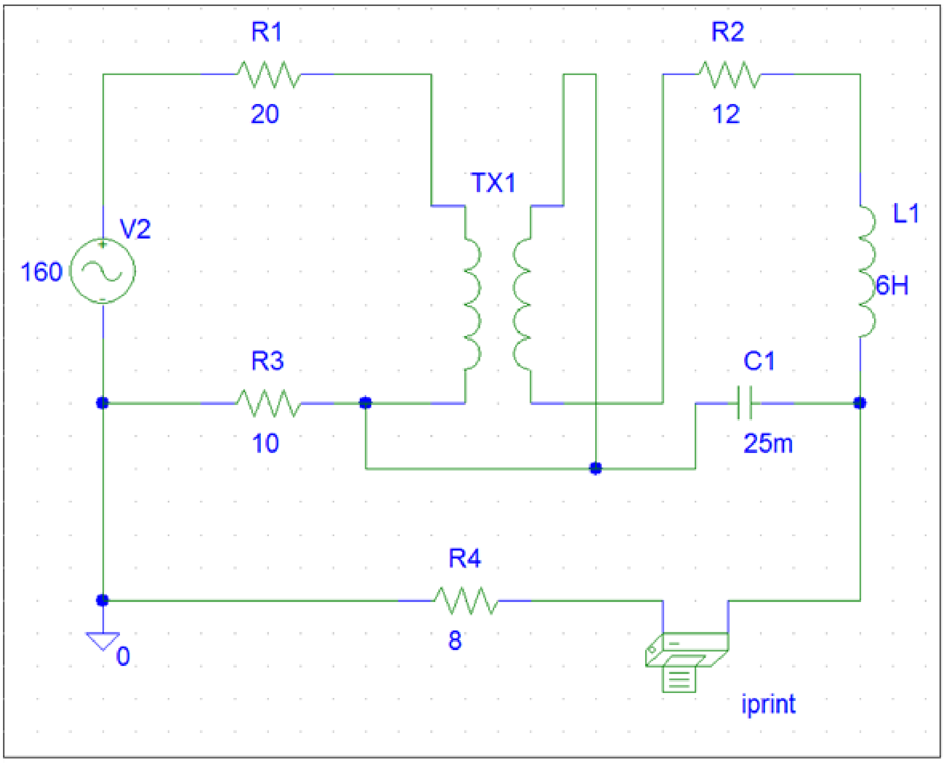
Step 2 of 6



Step 3 of 6

Step 2:

After making all connections by using *PSpice* given circuit is as below



Step 4 of 6

Step 3:

Double click on VAC voltage source and set $\mathrm{DC}=0\mathrm{V},\ \ \mathrm{ACMAG}=160$ and $\mathrm{ACPHASE}=50$ and double click on IPRINT and set $\mathrm{AC}=\mathrm{yes},\ \ \mathrm{MAG}=\mathrm{yes}$ and $\mathrm{PHASE}=\mathrm{yes}$.

Step 5 of 6

Step 4:

$\omega = 4\ \mathrm{rad/sec}$

$2\pi f = 4$

$f = \dfrac{4}{2\pi}$

$f = 0.63662\ \mathrm{Hz}$

Go to analysis / set up and select AC sweep / linear and enter total points = 1, start frequency $=0.63662\ \mathrm{Hz}$ end frequency $=0.63662\ \mathrm{Hz}$.

Step 6 of 6

Step 5:

We select analysis / simulate it. The output file induces.

FREQ IM(V_PRINT1) IP(V_PRINT1)

6.366E-01 2.011E+00 6.851E+01